Macroscopic Ensemble Dynamics of the Stable Secular Manifold Propagator in 8D Phase Space

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Abstract

Traditional N-body integrations are bottlenecked by $\mathcal{O}(N^2)$ computational complexity and the necessity of highly refined procedural time-steps in chaotic regimes. The Stable Secular Manifold Propagator (SSMP) mitigates this in non-interacting discrete systems by executing $\mathcal{O}(1)$ analytical jumps along an 8-dimensional phase-space trace against a central potential. This paper evaluates the SSMP's capacity to model macroscopic ensemble dynamics—specifically the phase-mixing of heavily perturbed debris disks. Through a high-density Monte Carlo parameter sweep evaluating 10^6 independent trajectories, we demonstrate that the SSMP accurately captures the natural radial dispersion and kinetic ejection thresholds of highly noisy initial conditions. The engine preserves strict $\mathcal{O}(1)$ computational invariance per particle, offering a massive acceleration for statistical and macroscopic orbital mechanics without relying on procedural discretization.

1 Introduction

Simulating the macroscopic evolution of planetary debris disks or perturbed halos traditionally requires a choice: utilize computationally expensive N-body integrators, or rely on generalized fluid-dynamic approximations. The Stable Secular Manifold Propagator (SSMP) offers an alternative for systems dominated by a central potential. By treating discrete orbital evolution as an analytical geodesic on an 8D phase-space trace, the SSMP executes time-jumps in constant $\mathcal{O}(1)$ time per particle.

While the SSMP's precision has been validated for single-body trajectories, its utility in statistical mechanics and ensemble dynamics requires empirical validation. This study scales the SSMP to large particle ensembles ($N \geq 10,000$) to evaluate its macroscopic predictive capabilities, specifically observing how it handles highly chaotic initial conditions, natural phase-mixing, and trajectory ejection.

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2 The Discrete 8D State Space

Let the extended phase space of the system be denoted by $\mathcal{M}^8$. For a discrete system of N non-interacting particles orbiting a central attractor, the state of the i-th particle at continuous secular time τ is defined by an 8-dimensional coordinate vector:

$$\boldsymbol{Z}_i(\tau) = (Z_i^1, Z_i^2, \ldots, Z_i^8) \in \mathcal{M}^8 \tag{1}$$

Rather than evolving via procedural differential equations (e.g., Runge-Kutta), the SSMP utilizes an analytical jump process. The evolution is defined by a deterministic flow map Φ_τ:

$$\boldsymbol{Z}_i(\tau) = \Phi_\tau(\boldsymbol{Z}_i(0)) \tag{2}$$

3 Macroscopic Ensemble Diagnostics

To evaluate the macroscopic behavior of the SSMP under massive initial perturbations, we treat the N particles as an empirical probability distribution. Assuming uniform particle mass, the radial density profile of the ensemble at time τ is analyzed.

To quantify the macroscopic phase-mixing (dispersion) of the system without succumbing to the curse of dimensionality, we compute the 1st Wasserstein distance (W_1) on the radial phase-space projection of the ensemble. We compare the evolved radial probability distribution to an unperturbed circular secular state at $\tau = T_{\text{sec}}$:

$$W_1(r_{\text{evolved}}, r_{\text{secular}}) = \inf_{\pi \in \Pi} \int_{\mathbb{R}\times\mathbb{R}} |x - y| d\pi(x, y) \tag{3}$$

This metric captures the radial broadening (secular phase-mixing) of the debris disk driven by the variance in the initial orbital elements.

4 Experimental Methodology: Monte Carlo Validation

We simulate a macroscopic collisionless system composed of $N = 10,000$ test particles. The baseline initial state is generated on a stable circular manifold. To construct the stress test, we systematically inject isotropic Gaussian noise into the Cartesian coordinates:

$$\boldsymbol{Z}_i(0) = \boldsymbol{Z}_{\text{circular},i} + \boldsymbol{\eta}_i, \quad \boldsymbol{\eta}_i \sim \mathcal{N}(0, \sigma_{\text{noise}}) \tag{4}$$

The Monte Carlo framework evaluates the temporal evolution of this noisy ensemble over a secular time horizon $\tau = 1000$ units. The noise amplitude σ_{noise} is swept incrementally from 1% to 95% to observe the boundary limits of the analytical engine. The simulation tracks both the radial Wasserstein dispersion (W_1) and the survival rate (operationally defined as maintaining $r < 50$ at $\tau = 1000$, which for initially near-circular orbits corresponds to remaining in a bound state, $e < 1$).

5 Results and Discussion

The simulation successfully processed 10^6 independent trajectories (100 distinct noise intervals) in approximately 12 seconds on standard consumer hardware. The macroscopic behavior of the ensemble is summarized in Figure 1.

Figure 1: Macroscopic phase-mixing and survival rates of the ensemble as a function of injected kinetic chaos. The solid cyan line represents the radial dispersion (W_1 distance), while the dashed red line tracks the bound survival fraction (particles remaining within $r < 50$).

5.1 Computational Invariance

The most striking empirical result is the validation of the engine's asymptotic temporal scaling. Standard procedural integrators experience exponential computational decay in high-noise regimes due to the necessity of vanishingly small time-steps required to resolve highly eccentric orbits. In contrast, the SSMP evaluates the $\tau = 1000$ analytical jump in strictly constant time $\mathcal{O}(1)$ per particle against the central potential, entirely independent of the injected noise amplitude. For comparison, a standard adaptive-step integrator (e.g., IAS15) would require $\mathcal{O}(10^3 - 10^4)$ time-steps per trajectory for comparable accuracy in high-eccentricity regimes, resulting in $\mathcal{O}(10^9 - 10^{10})$ force evaluations to achieve what the SSMP resolved in seconds.

5.2 Natural Phase-Mixing and Dispersion

As observed in the fluid dispersion metric (the solid cyan line in Figure 1), the macroscopic width of the orbital manifold accurately scales with the injected kinetic noise. The engine does not artificially force the perturbed states back to the baseline circular manifold; rather, it correctly propagates the altered eccentricities and semi-major axes. This results in the natural, physically expected phase-mixing of the debris disk, where higher initial kinetic noise translates smoothly into a wider steady-state radial distribution.

5.3 The Escape Threshold and Ejection

The survival rate metric reveals a distinct physical phase transition. Up to an injection threshold of roughly 15% kinetic noise, the ensemble is strictly bounded, maintaining a near 100% survival rate. Beyond this threshold, the kinetic variance is sufficient to kick a statistically significant fraction of the particles across the $e \geq 1$ boundary layer.

The ensemble begins to naturally "evaporate" into unbound escape trajectories. By 95% noise, nearly 80% of the ensemble has been ejected. Crucially, the SSMP identifies and resolves these transitions analytically, cleanly separating bound states from hyperbolic escapes without crashing or requiring the adaptive step-size corrections that typically stall standard numerical solvers.

6 Conclusion

By treating orbital mechanics as an analytical jump along an 8D phase-space trace, the Stable Secular Manifold Propagator (SSMP) offers a massive computational acceleration for macroscopic, non-interacting ensembles. Our empirical Monte Carlo sweep demonstrates that the engine accurately scales to large particle counts ($N = 10,000$), correctly capturing the natural radial dispersion and phase-mixing of highly perturbed systems. Furthermore, it natively identifies kinetic ejection thresholds ($e \geq 1$) while maintaining strict $\mathcal{O}(1)$ computational time per particle. This positions the SSMP as a highly efficient, procedural-free tool for generating rapid statistical predictions of debris disks and planetary rings.

Data Availability

The core $\mathcal{O}(1)$ *Stable Secular Manifold Propagator* (SSMP), alongside the Monte Carlo stress-testing scripts used to generate the macroscopic dispersion metrics in this study, are open-source. The codebase and execution environments are publicly available in the GitHub repository: `https://github.com/SauliKalkaja/Symplectic-8D-NBody-Manifold`.